



LOMBARD ODIER
INVESTMENT MANAGERS

Empirical Methods in Finance

Class 4 - How to Use Autoregressive Models in Finance

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March 7, 2026

Lombard Odier Investment Management

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Basics of ARMA models

What is the impact of an MA noise?

Time series effect, i.e. the disturbances are not what you think but rather $\epsilon_t = \rho\eta_{t-1} + \eta_t$. In such as case, the true model is

$$r_t = \beta r_{m,t} + \rho\eta_{t-1} + \eta_t$$

. This leads to

$$\hat{\beta} = \frac{\sum_t r_{m,t}(\beta r_{m,t} + \rho\eta_{t-1} + \eta_t)}{\sum_t r_{m,t}^2} = \beta + \rho \frac{\sum_t r_{m,t}\eta_{t-1}}{\sum_t r_{m,t}^2} + \frac{\sum_t r_{m,t}\eta_t}{\sum_t r_{m,t}^2}$$

.
Taking expectations yields $\hat{\beta} = \beta + \rho \frac{\text{Cov}(\eta_{t-1}, r_{m,t})}{V[r_{m,t}]}$.

⇒ The last model is a *time series* model and it incorporates an MA(1) disturbance.

Founding element: white noises.

ϵ_t follows a white noise process if

- i.i.d.
- Gaussian
- Expectation 0 and variance σ_ϵ .

Time series models are combinations of white noise with path dependent components.

In financial modeling, white noises are very important: Black-Scholes model is a white noise model augmented with a drift.

Stationarity

Two different definitions:

1. Strict stationarity: Let X_t be a timeseries process. $\forall h$ X_t and X_{t+h} have the *same* distribution. Hard to prove, hard to test.
2. Second order stationarity: Let X_t be a timeseries process. Then if
 - $\mathbb{E}[X_t] = \mu < \infty$
 - $\mathbb{V}[X_t] = \sigma^2 < \infty$
 - $\text{Cov}(X_t, X_{t+h}) = f(h) < \infty$

X_t is said to be second order stationary.

And then the Wold theorem: any second order stationary process X_t can always be represented as an infinite sum of past and present shocks:

$$X_t = \sum_{i=0}^{\infty} \psi_i \epsilon_{t-i} + \kappa_t,$$

where $\psi_i \in \mathbb{R}$, $\psi_0 = 1$, $\sum_{i=0}^{\infty} \psi_i^2 < \infty$ and ϵ_t is a white noise. κ_t is a function of t and is not stochastic.

Three classic timeseries models:

- Autoregressive models (AR(p)) models:

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t$$

- Moving Average models (MA(q)) models:

$$X_t = \sum_{i=1}^q \psi_i \epsilon_{t-i} + \epsilon_t$$

- Autoregressive models (ARMA(p,q)) models:

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \sum_{i=1}^q \psi_i \epsilon_{t-i} + \epsilon_t$$

Identification

How to identify an MA model from an AR model?

Before performing an estimation, their "autocorrelogram" are informative: autocorrelation between "lags" is obtained from the following formula:

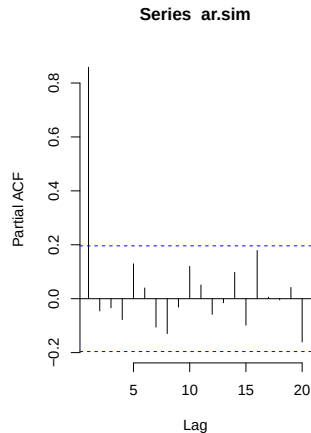
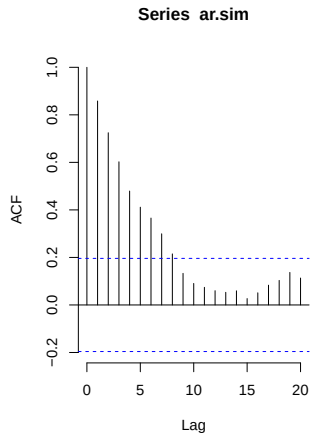
$$\gamma_h = \frac{\text{cov}(X_t, X_{t+h})}{V(X_t)}$$

- In the case of an MA(q) process, the autocorrelation function is different from 0 up to $h = q$ and 0 afterward.
- In the case of an AR(p) process, the autocorrelation function always different from zero and decays slowly to zero as h increases.

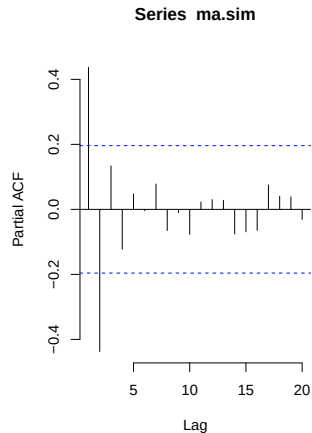
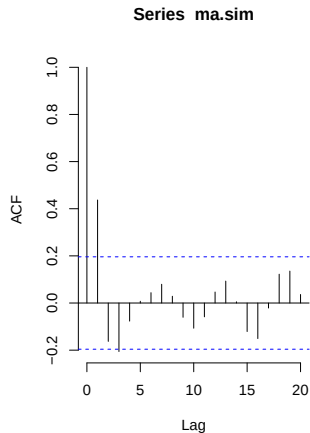
This result can be easily explained with the Wold representation of an AR process.

Partial autocorrelation can be computed from a linear regression of X_t on its lags and yields exactly the opposite image.

ACF/PACF AR(1) process



ACF/PACF MA(1) process



Both timeseries models are no longer iid. How to deal with that? By using conditional densities, we can estimate their parameters by maximum likelihood:

$$f(X_1, X_2, X_3, \dots, X_n) = f(X_1)f(X_2|X_1)f(X_3|X_2, X_1) \times \dots \times f(X_n|X_1, \dots, X_{n-1})$$

Taking the log of this expression yields the following log likelihood:

$$\log L = \log f(X_1) + \log f(X_2|X_1) + \log f(X_3|X_2, X_1) + \dots + \log f(X_n|X_1, \dots, X_{n-1})$$

The loglikelihood estimates are obtained by maximizing this expression. The conditional densities are obtained by using the usual change in variable theorem.

Important remark: the MA process is path dependent and the ϵ_t are unobservable. We can only know its likelihood function for a given set of parameters. It therefore needs to be maximized numerically. Once estimated it yields a timeseries of individual shocks as a by-product.

The case of an AR process is much simpler given it is a linear model in observed variables (past realizations of X_t): OLS estimates can be used and coincide with ML estimates.

How to chose p and q?

Three information criterions can be used:

- Akkaike criterion: $AIC(p, q) = 2(p + q) - 2 \log L$
- Schwartz criterion: $BIC(p, q) = 2(p + q) \log n - 2 \log L$
- Hanan and Quinn criterion: $HQ(p, q) = 2(p + q) \log \log n - 2 \log L$,

with n the number of observations. The right p and q should minimize one of these three information criterion.

A first example : back with the Taylor rule

The model:

$$r_t = \alpha + \beta_\pi \pi_t + \beta_g g_t + \sigma \epsilon_t,$$

with $\epsilon_t \sim N(0, 1)$.

When the Central Bank adjust its target rate smoothly, it gives rise to an MA effect in ϵ_t which becomes:

$$\epsilon_t = \rho \eta_{t-1} + \eta_t,$$

with $\eta_t \sim N(0, 1)$.

Using the change in variable theorem, we obtain the following log-likelihood function:

$$\log L(\theta) = \sum_{t=1}^n \log f \left(\frac{r_t - (\beta_\pi \pi_t + \beta_g g_t - \sigma \rho \eta_{t-1})}{\sigma} \right) \times \frac{1}{\sigma}$$

⇒ Numerical optimization: this is the way.

Introduction to Optimization

No analytical solutions?

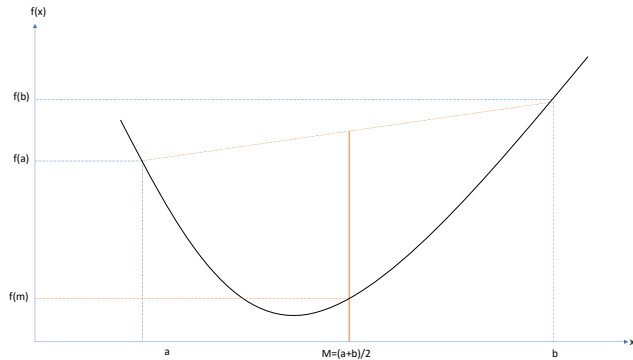
Most of the time, you will NOT find an analytical solution to the ML max program. How can we deal with it?
Numerical optimization: when things get complicated.

Assume you need to minimize a function $h(x)$ of one variable, with $x \in [a, b]$ but can't do it through a closed form formula. Two solutions:

- Grid search: slide and dice $[a, b]$ and compute the function for each node. Retain x that yielded the max value.
- Dichotomy method: if the function has a unique max:
 - compute $m = (a + b)/2$
 - if $f(b) > f(a) > f(m)$ then the solution is between m and a and b become m
 - if $f(a) > f(b) > f(m)$ then the solution is between m and b and a become m
 - Continue until $\text{abs}(a - b) < \epsilon$.

Works only if $f''(x) \leq 0$

An Illustration of the Dichotomy



No analytical solutions?

What happens when more than one parameter needs to be estimated? numerical optimization.

Two solutions:

- be lazy, use 'optim' in R. Eventually, you'll chose this solution.
- be a good student and understand the basics of the Newton Raphson optimization method.

A crash introduction to Newton Raphson

All gradient based optimization methods are based on the same idea: create a sequence of estimators that converges towards the 'right' solution.

Let θ be the sequence of parameters we need to estimate and θ_i the i^{th} step of the optimization.

We want to create a sequence such that

$$\theta_{i+1} = \theta_i + \lambda_i \Delta_i$$

with

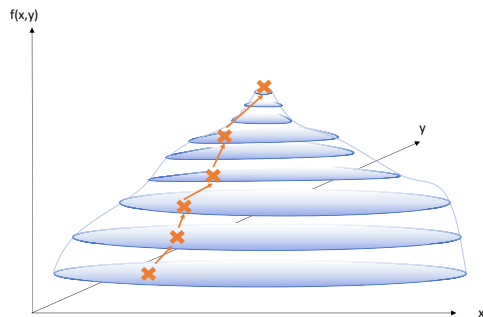
- λ_i is the size of the step to be done
- Δ_i the direction of the step.

Gradient-based methods use $\Delta_i = W_i G_i$, where G_i is the gradient matrix

$$G_i = \left[\frac{\partial LL(\theta_i)}{\partial \theta_i} \right]$$

and W_i a definitive positive matrix.

An Illustration of the Newton Raphson Method



A crash introduction to Newton Raphson

Newton-Raphson:

Assume you want to maximize $f : \mathbb{R}^k \rightarrow \mathbb{R}$, a function that can be differentiated twice and whose derivatives are continuous.

A Taylor approximation of $f(\cdot)$ yields:

$$f(x + h) = f(x) + G^\top h + \frac{1}{2} h^\top H h$$

, with H the Hessian matrix of f .

Differentiating this expression and solving it for zero yield the optimal h :

$$\partial_h f(x + h) = G + Hh = 0 \Rightarrow h = -H^{-1}G$$

.

This means: around θ_i , the best step is $-H_i^{-1}$ to be combined with $\Delta_i = G_i$

A crash introduction to Newton Raphson

Newton-Raphson's programming:

1. Start from a given θ_0
2. Compute G_0 and H_0
3. Compute $\lambda_0 = 1$ and $\Delta_1 = -H_0^{-1}G_0$
4. obtain θ_1
5. test that $\|G\| < \epsilon$, if not start again with θ_1 as a starting point.

The computation of H can be tedious, even numerically. Trick of the day: use the BHHH approximation:

$$-H = GG^\top,$$

as

$$-\mathbb{E} [\partial_\theta^2 LL(\theta_i)] = \mathbb{E} [\partial_\theta LL(\theta)^\top \partial_\theta LL(\theta)]$$

. The sequence becomes

$$\theta_{i+1} = \theta_i + (GG^\top)^{-1}G$$

What Can We Do With ARMA models?

What Can We Do With ARMA models?

In truth? Not much.

But financial econometrics came up with two main usages:

- First, checking once more than returns are *not persistent* on average, granting credency to the Efficient Market Hypothesis.
- Second, modeling volatility (more exactly *variance*) as volatility is a persistent phenomenon.
- Third, modeling rates as rates are nearly integrated time series.

Basically

- ARMA models are widely used in economics and finance to give a dynamic shape to anything persistent enough (GDP growth, inflation, interest rates, volatility...).
- But not returns: persistent returns would open the door systematic strategies able to arbitrage this persistence away $\Rightarrow$ No persistence in returns (and no need to check, you won't find any).

The Efficient Market Hypothesis

The efficient-market hypothesis (EMH):

- is a hypothesis in financial economics that states that asset prices reflect all available information.
- A direct implication is that it is impossible to "beat the market" consistently on a risk-adjusted basis since market prices should only react to new information.
- A third implication: markets follow a "random walk", basically markets are unpredictable white noise.

Historical reference: Fama, Eugene (1970). "Efficient Capital Markets: A Review of Theory and Empirical Work". *Journal of Finance*. 25 (2): 383–417.

All of that can be written down as follows:

$$r_t = \phi_0 + \sigma \epsilon_t \quad (1)$$

$$\epsilon_t \sim N(0, 1). \quad (2)$$

The Efficient Market Hypothesis

One key property of ARMA models: their known persistence structure.

AR(1) model: $x_t = \phi_0 + \phi_1 x_{t-1} + \sigma \epsilon_t$, with $\epsilon_t \sim \text{i.i.d. } N(0, 1)$.

$$\text{Cov}(x_t, x_{t-1}) = \text{Cov}(\phi_0 + \phi_1 x_{t-1}, x_{t-1}) \quad (3)$$

$$= \phi_1 \text{Cov}(x_{t-1}, x_{t-1}) \quad (4)$$

$$= \phi_1 \sigma_x^2, \quad (5)$$

where $\sigma_x^2 = \mathbb{V}[x_t] = \frac{\sigma^2}{1 - \phi_1^2}$.

$$\text{Cov}(x_t, x_{t-2}) = \text{Cov}(\phi_0 + \phi_1(\phi_0 + \phi_1 x_{t-1}), x_{t-1}) \quad (6)$$

$$= \phi_1^2 \sigma_x^2. \quad (7)$$

From which it is easy to show that:

$$\text{Cov}(x_t, x_{t-h}) = \phi_1^h \sigma_x^2. \quad (8)$$

The Efficient Market Hypothesis

One key property of ARMA models: their known persistence structure.

MA(1) model: $x_t = \phi_0 + \phi_1\sigma\epsilon_{t-1} + \sigma\epsilon_t$, with $\epsilon_t \sim \text{i.i.d. } N(0, 1)$.

$$\text{Cov}(x_t, x_{t-1}) = \text{Cov}(\phi_0 + \phi_1\sigma\epsilon_{t-1} + \sigma\epsilon_t, \phi_0 + \phi_1\sigma\epsilon_{t-2} + \sigma\epsilon_{t-1}) \quad (9)$$

$$= \phi_1\sigma^2 \text{Cov}(\epsilon_{t-1}, \epsilon_{t-1}) \quad (10)$$

$$= \phi_1\sigma^2. \quad (11)$$

$$\text{Cov}(x_t, x_{t-2}) = \text{Cov}(\phi_0 + \phi_1\sigma\epsilon_{t-1} + \sigma\epsilon_t, \phi_0 + \phi_1\sigma\epsilon_{t-3} + \sigma\epsilon_{t-2}) \quad (12)$$

$$= 0. \quad (13)$$

From which it is easy to show that:

$$\text{Cov}(x_t, x_{t-h}) = 0 \forall h > 1. \quad (14)$$

The Efficient Market Hypothesis

An easy test to check for the Efficient Market Hypothesis:

$$r_t = \phi_0 + \phi_1 r_{t-1} + \sigma \epsilon_t \quad (15)$$

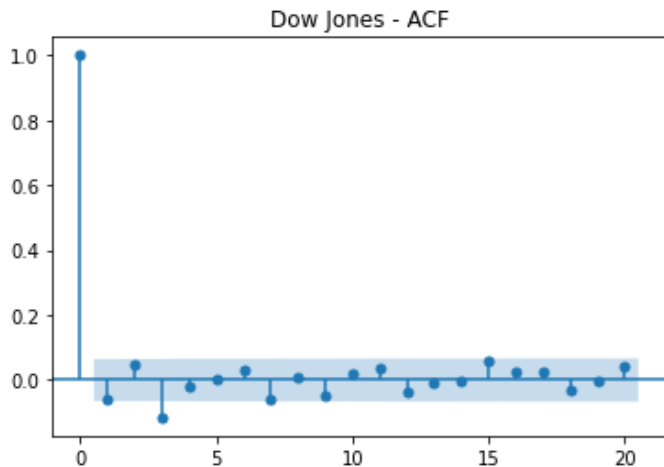
$$\epsilon_t \sim N(0, 1). \quad (16)$$

Two tests:

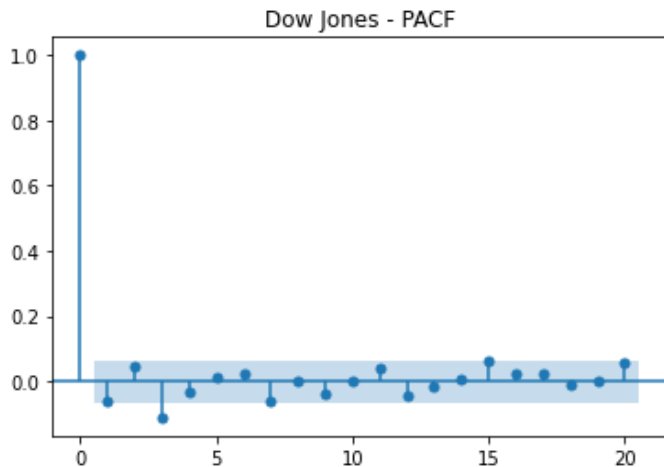
- First, look for the ACF/PACF of financial returns. If an ARMA structure exists, then it should be visible.
- Second, build a genuine AR(1) model as above and estimate it by maximum likelihood and check for parameter significance.

Do both using the sample of weekly returns on the Dow Jones for the 2002-2019 period.

The Efficient Market Hypothesis



The Efficient Market Hypothesis



The Variance Modeling Approach

Modelling volatility is an essential task for financial econometrician.

Conditional volatility defines as

$$\mathbb{V} \left[r_t | \underline{r_{t-1}} \right]$$

This probability concepts needs to be approached with an estimate that can then be modelled.

A first estimate: realized variance

Intraday data makes it possible to derive high frequency volatility estimates, as done by Barndorff-Nielsen and Shephard (2002).

Assuming a trading day t can be split into M subperiods, intraday returns are $r_{t,j} = \log p_{t,j} - \log p_{t,j-1}$ for $j = 2, \dots, M$.

The easiest of these estimators: the Realized variance is obtained as

$$RV_{t,M} = \sum_{j=1}^M r_{t,j}^2$$

The Variance Modeling Approach

Any process S_t can be represented the following way:

$$\frac{dS_t}{S_t} = \underbrace{\mu dt}_{\text{Deterministic part}} + \underbrace{\sigma dW_t}_{\text{Diffusive part}} + \underbrace{J_t dN_t}_{\text{Jump part}} \quad (17)$$

Most of the time series models' approach are actually mixing the diffusive and the jump part of the processes:

$$r_t = \mu + \sigma_t \epsilon_t, \quad (18)$$

$$\epsilon_t \sim IID(\text{Possibly skewed and leptokurtic}) \quad (19)$$

⇒ The Realized Variance estimate produces variance estimates inclusive of the jump part.

The Variance Modeling Approach

The solution: the joint probability of a jump over two consecutive trading intervals is smaller.

A bipower variation estimate is

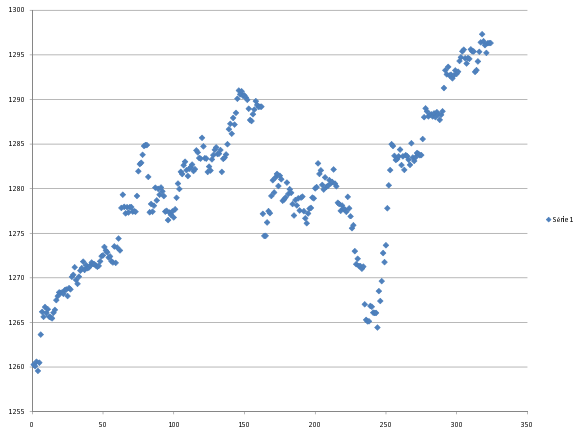
$$BPV(t, M) = \sum_{j=2}^{M-1} |r_{t,j-1}| \times |r_{t,j}| \quad (20)$$

The difference between both measures often used as a proxy for the jump activity in a trading day. On bipower variation, main reference is:

- Barndorff-Nielsen and Shephard (2004), Power and bipower variation with stochastic volatility and jumps, Journal of Financial Econometrics 2:1-48
- For an extensive use of it for empirical finance, see Jumps, CoJumps and Macro Announcements (with Jérôme Lahaye and Chris Neely).

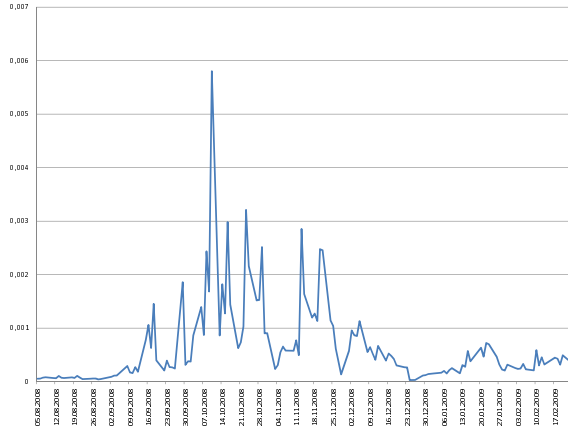
The Variance Modeling Approach

Data snapshot: high frequency data from 05.08.2008 to 08.08.2008.



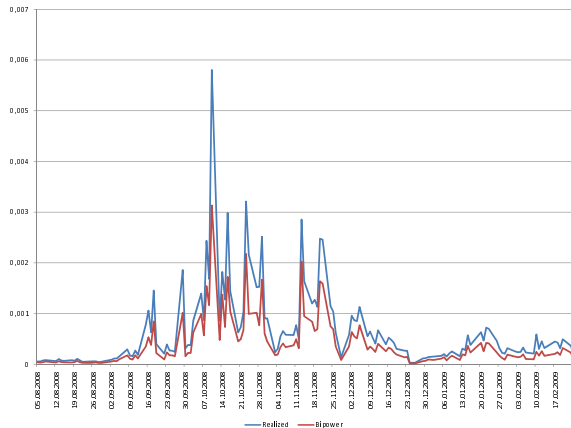
The Variance Modeling Approach

Realized estimates from 05.08.2008 to 08.08.2008.



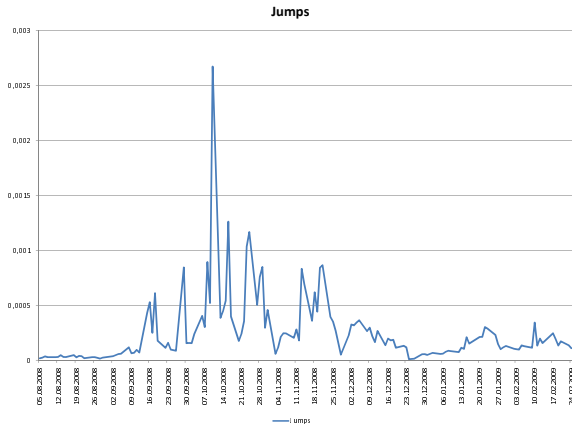
The Variance Modeling Approach

Bipower estimates from 05.08.2008 to 08.08.2008.



The Variance Modeling Approach

Jump activity from 05.08.2008 to 08.08.2008.



The Variance Modeling Approach

An interesting issue: micro-structure noise

The intuition would say to use data at the lowest frequency possible, so as to be as close as possible to the limit of the estimator.

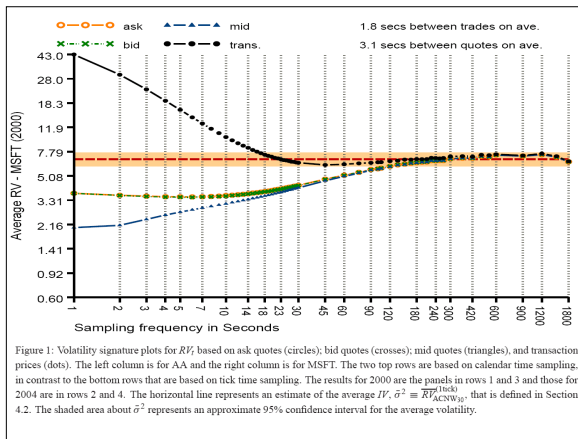
However, market quotes are affected by specific phenomenon labelled as "microstructure noise", related to the organization of the market, such as:

- infrequent trading
- seasonality
- price manipulation
- ...

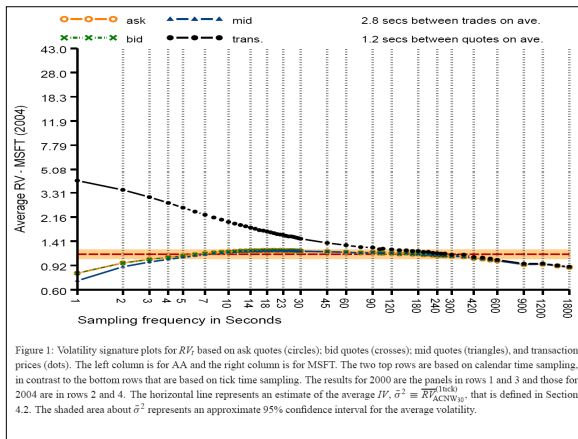
The higher the frequency used to compute both formula and the higher the likelihood that the estimator is affected by a micro structure noise.

Signature plot: plot the average volatility w.r.t. the sampling frequency. See Hansen and Lund (2004). $\Rightarrow$ The estimator is said to be free from micro-structure noise when volatility converges to a certain level.

The Variance Modeling Approach



The Variance Modeling Approach



The Variance Modeling Approach

What can we do with that?

1. Increasing our understanding about the jump activity in financial returns. Is it weak? Is it strong? Does prices move only by jumps?
2. Uncovering of the jump to volatility channel by Aït-Sahalia, Cacho-Diaz and Laeven (2011), Carr and Wu (2011), and Fulop, Li and Yu (2011).
3. Risk management: forecasting the density of returns with or without this jump component? See Maheu and McCurdy (2009).
4. Estimating continuous time diffusion easily.
5. ...

The Variance Modeling Approach

Estimation of the Heston volatility dynamics

In the Heston framework: Cox-Ingersoll-Ross process for the volatility:

$$dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dW_t, \quad (21)$$

Conditional distribution of the volatility is a non-centered gamma distribution, up to a change of scale.

$$\rho = \exp(-\kappa), \quad \delta = \frac{2\kappa\theta}{\sigma^2}, \quad c_{\Delta t} = \frac{\sigma^2}{2\theta}(1 - \exp(-\theta\Delta t)) \quad (22)$$

Then:

$$\mathbb{E}[V_{t+\Delta t}|V_t] = c_{\Delta t}\delta + \rho^{\Delta t}V_t \quad (23)$$

$$\mathbb{V}[V_{t+\Delta t}|V_t] = c_{\Delta t}^2\delta + 2\rho^{\Delta t}c_{\Delta t}V_t \quad (24)$$

Thus, it is possible to estimate the CIR dynamic for the volatility from the following discrete time model:

$$V_{t+\Delta t} = c_{\Delta t}\delta + \rho^{\Delta t}V_t + u_{t,\Delta t} \quad (25)$$

$$\mathbb{E}[u_{t,\Delta t}|V_t] = 0, \quad \mathbb{E}[u_{t,\Delta t}^2|V_t] = c_{\Delta t}^2\delta + 2\rho^{\Delta t}c_{\Delta t}V_t \quad (26)$$

Can be estimated in a consistent way by Generalized Least Squares (GLS). Use jointly the method of moments and GLS to estimate the whole parameters.

The Variance Modeling Approach

Estimates

Using method of moment estimates, we obtain:

	κ	θ	σ	ρ
Realized	121	16%	0.1%	-48%
Bipower	117	10%	0.07%	-24%

Clearly:

- depending on the vol measure you are using results are different
- the elimination of jumps reduces the vol of vol parameter
- also an impact on the persistence parameter

The Variance Modeling Approach

μ	α	β	σ	ρ	λ	μ_j	σ_j	μ_v	ρ_j
Jiang and Knight (sample: 1990-1999)									
0.056	0.783	0.230	0.820	-0.273					
(0.017)	(0.023)	(0.101)	(0.201)	(0.114)					
Chacko and Viceira (sample: 1990-1999)									
0.1406	0.0255	16.6997	4.7148	-0.4125					
(0.0573)	(0.0039)	(4.5364)	(0.8860)	(0.2269)					
Chacko and Viceira (sample: 1980-2000)									
0.1121	0.0330	14.2819	5.1925	-0.6289					
(0.0379)	(0.0021)	(3.2752)	(0.4722)	(0.2097)					

Source: Rockinger, M. And Semenova, M. Estimation of Jump-Diffusion Processes via Empirical Research Characteristic Functions. Paper N°150 June 2005.

The Variance Modeling Approach

Corsi (2009) postulates that this Realized Variance can be modeled using a "component" AR model defined as follows:

$$\log(RV_t) = \omega + \phi_1 \log(RV_{t-1}) + \phi_2 \log(RV_{t-5:t-1}) \quad (27)$$

$$+ \phi_3 \log(RV_{t-22:t-1}) + \sigma u_t. \quad (28)$$

Called the Heterogenous AutoRegressive model (HAR).

The estimation via maximum likelihood is straightforward and yields the following estimated values for the S&P500:

$$\omega = -1.391 \quad (29)$$

$$\phi_1 = 0.206 \quad (30)$$

$$\phi_2 = 0.391 \quad (31)$$

$$\phi_3 = 0.264 \quad (32)$$

This is actually inspired from other volatility models: the GARCH models.

Using "data_rv.csv" estimate the parameter of an HAR model by maximum likelihood.

Modelling Rates

The Vasicek Model starts from an AR(1) model for short term rates (rates with the smallest possible maturity, e.g. one day)

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t \quad (33)$$

This equation can be discretized as follows:

$$r_t - r_{t-1} = \kappa(\theta - r_{t-1}) \times (t + 1 - t) + \sigma \epsilon_t \quad (34)$$

$$\Leftrightarrow r_t = \kappa\theta + (1 - \kappa)r_{t-1} + \sigma\epsilon_t \quad (35)$$

$$\Leftrightarrow r_t = \rho_0 + \rho_1 r_{t-1} + \sigma\epsilon_t \quad (36)$$

ϵ_t follows a Gaussian distribution (0,1) under the historical distribution. The risk neutral distribution is obtained as follows:

$$\epsilon_t = \underbrace{\Delta_0 + \Delta_1 r_{t-1}}_{\text{Risk Premia}} + \epsilon_t^Q \quad (37)$$

Modelling Rates

Injecting it in the dynamics of rates equation yields the dynamics of r_t under the risk neutral distribution

$$r_t = \rho_0 + \rho_1 r_{t-1} + \sigma(\Delta_0 + \Delta_1 r_{t-1} + \sigma \epsilon_t^Q) \quad (38)$$

$$\Leftrightarrow r_t = \underbrace{\rho_0 + \sigma \Delta_0}_{\rho_0^*} + \underbrace{\rho_1 + \sigma \Delta_1}_{\rho_1^*} r_{t-1} + \sigma \epsilon_t^Q \quad (39)$$

$$\Leftrightarrow r_t = \rho_0^* + \rho_1^* r_{t-1} + \sigma \epsilon_t^Q \quad (40)$$

Under the risk neutral distribution, short term rates are still evolving as an autoregressive model.

The price of a bond paying 1 Dollar at time T on trading day t is obtained as

$$P(t, T) = \mathbb{E}^Q \left[e^{-\sum_{s=t+1}^T r_s} | r_t \right]. \quad (41)$$

r_t here is pro-rated rates, meaning annualized rates divided by 12 in the case of monthly steps between t and $t + 1$.

The model here is affine in its risk factors, which implies the solution to the price of a bond is

$$P(t, T) = e^{A(t, T) + B(t, T) r_t}. \quad (42)$$

The model here is affine in its risk factors, which implies the solution to the price of a bond is

$$P(t, T) = e^{A(t, T) + B(t, T)r_t}. \quad (43)$$

$A(t, T)$ and $B(t, T)$ are functions of the difference between T and t , they do not rely on t or T specifically. Interestingly, when $t = T$, meaning the bond is maturing being therefore worth 1, we have

$$P(T, T) = e^{A(T, T) + B(T, T)r_T} = 1 \Rightarrow B(T, T) = A(T, T) = 0 \quad (44)$$

We do not know yet what $A(\cdot)$ and $B(\cdot)$ look like, but we can derive a relation between $A(t, T)$, $B(t, T)$, $A(t+1, T)$ and $B(t, T)$. Such a relation is called a "Ricatti equation".

$$P(t, T) = \mathbb{E}^Q \left[e^{-\sum_{s=t+1}^T r_s} | r_t \right] \quad (45)$$

$$= \mathbb{E}^Q \left[e^{-r_{t+1} - \sum_{s=t+2}^T r_s} | r_t \right] \quad (46)$$

$$= \mathbb{E}^Q \left[e^{-r_{t+1}} \mathbb{E}^Q \left[e^{-\sum_{s=t+2}^T r_s} | r_{t+1} \right] | r_t \right] \quad (47)$$

$$= \mathbb{E}^Q \left[e^{-r_{t+1}} P(t+1, T) | r_t \right] \quad (48)$$

$$= \mathbb{E}^Q \left[e^{-r_{t+1} + A(t+1, T) + B(t+1, T)r_{t+1}} | r_t \right] \quad (49)$$

$$= \mathbb{E}^Q \left[e^{A(t+1, T) + (B(t+1, T) - 1)(\phi_0 + \phi_1 r_t + \sigma \epsilon_{t+1})} | r_t \right] \quad (50)$$

$$= e^{A(t+1, T) + (B(t+1, T) - 1)(\phi_0 + \phi_1 r_t)} \mathbb{E}^Q \left[e^{(B(t+1, T) - 1)\sigma \epsilon_{t+1}} | r_t \right] \quad (51)$$

$$P(t, T) = e^{A(t+1, T) + (B(t+1, T) - 1)(\phi_0 + \phi_1 r_t)} \mathbb{E}^Q \left[e^{(B(t+1, T) - 1)\sigma \epsilon_{t+1}} | r_t \right] \quad (52)$$

We know that

$$\epsilon_t \sim N(0, 1) \Rightarrow \mathbb{E}[e^{w \epsilon_t}] = e^{-\frac{1}{2} w^2} \quad (53)$$

Therefore:

$$P(t, T) = e^{A(t+1, T) + (B(t+1, T) - 1)(\phi_0 + \phi_1 r_t) + \frac{1}{2} (B(t+1, T) - 1)^2 \sigma^2} \quad (54)$$

Which implies

$$A(t, T) = A(t+1, T) + \phi_0 (B(t+1, T) - 1) - \frac{1}{2} (B(t+1, T) - 1)^2 \sigma^2 \quad (55)$$

$$B(t, T) = \phi_1 (B(t+1, T) - 1). \quad (56)$$

Applications

Dataset 1 - Taylor rule with recursive MA(1) noise

Model:

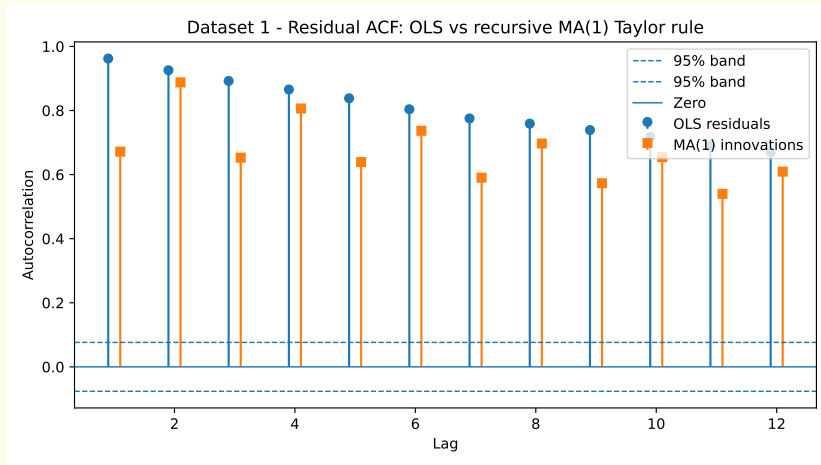
$$i_t = \alpha + \beta_\pi \pi_t + \beta_y gap_t + u_t, \quad u_t = \varepsilon_t + \theta \varepsilon_{t-1}.$$

Recursive innovation:

$$\varepsilon_t = i_t - \alpha - \beta_\pi \pi_t - \beta_y gap_t - \theta \varepsilon_{t-1}.$$

Parameter	OLS estimate	OLS std. err.	MA(1) MLE estimate	MA(1) MLE std. err.
Intercept	1.5031	0.1802	1.5779	0.2320
Inflation coefficient	0.8914	0.0358	0.8706	0.0307
Output gap coefficient	0.2201	0.0512	0.2096	0.0569
MA(1) coefficient			0.8099	0.0150
Innovation sd	2.6715		1.6096	0.0375

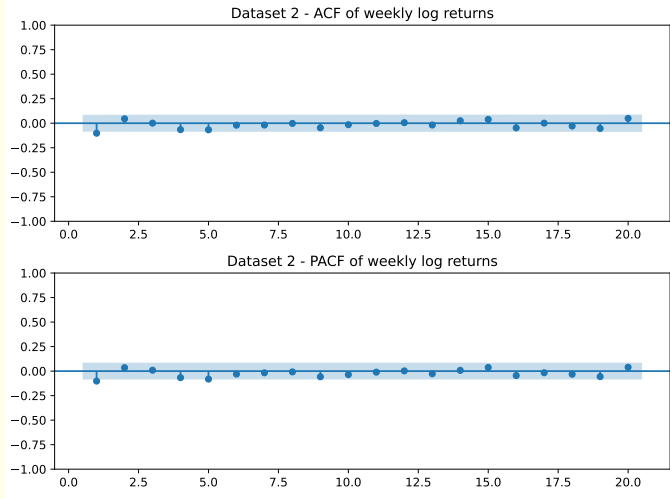
Dataset 1 - Residual diagnostics



Dataset 2 - Weekly return autocorrelation tests

Lag	Ljung-Box stat	Ljung-Box p-value	Box-Pierce stat	Box-Pierce p-value
5	11.1058	0.0493	11.0089	0.0512
10	12.7772	0.2364	12.6468	0.2441
20	18.6559	0.5443	18.3062	0.5672

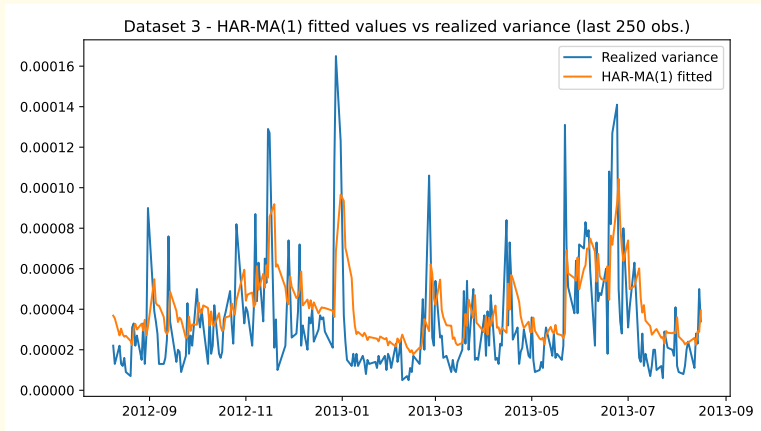
Dataset 2 - ACF and PACF of weekly returns



Dataset 3 - HAR-MA(1) parameter estimates

Parameter	Estimate	Std. err.	t-stat	p-value
Intercept	9.473e-06	5.745e-06	1.6489	0.0992
Lag 1 RV	0.2787	0.0272	10.2599	0.000e+00
Lag 5 average RV	0.4610	0.0198	23.2917	0.000e+00
Lag 20 average RV	0.1735	0.0102	16.9402	0.000e+00
MA(1) coefficient	-4.010e-15	0.0241	-1.661e-13	1.0000
Innovation sd	1.557e-04	2.587e-07	601.7757	0.000e+00

Dataset 3 - Fitted values



- Dataset 1: the Taylor rule is estimated with a recursive MA(1) likelihood and OPG standard errors built from numerical scores.
- Dataset 2: weekly S&P500 returns show little robust autocorrelation, broadly consistent with weak-form efficiency.
- Dataset 3: realized variance is strongly persistent across the daily, weekly, and monthly HAR horizons; the MA(1) term is comparatively small.